Chapter 4

Determinants

4.1 Why Determinants?

The previous chapter on matrices ended up with a set of conditions for a matrix to have an inverse. The easiest to write down is the condition that the $n \times n$ matrix A has an inverse if and only if there is no n vector for which $A\mathbf{x} = \mathbf{0}$ apart from $\mathbf{x} = \mathbf{0}$. This is fine as far as it goes, but not much practical use without guidance on finding out whether the condition is satisfied. So it would be nice to have a formula involving the components of the matrix A that told you directly whether A has an inverse. There is such a formula; it is called the determinant of A, written as $\det A$ or $|A|$, and the square matrix A has an inverse if and only if $\det A \neq 0$. This is the first part of Theorem 20 in this chapter. The determinant also fills another nasty gap. It would be very nice to have a formula for the inverse of A. There is one; it involves determinants. Theorem 20 in this chapter gives the formula. Determinants have further uses. I will be introducing the ideas of the eigenvalues and eigenvectors in chapter 5 which follows this one.. Determinants are used to find the eigenvalues of a matrix. You will meet positive and negative definite matrices. How do you find out if a matrix is positive or negative definite? Using the determinant of course.

This all sounds very good. The difficulty is that the formula for a determinant does not look at all nice when you come to write it down. In fact we never do write it down in full for anything bigger than a 3×3 matrix. We have a simple formula for the determinant of a 2×2 matrix, but the general formula is inductive, giving the determinant of an $n \times n$ matrix as a function of the determinants of n different $(n-1) \times (n-1)$ matrices. There some matrices for which the calculation of the determinant is straightforward, I demonstrate some of them. But in general calculating a determinant by hand is a very daunting task. Nevertheless, owing to the close links between determinants and solutions of simultaneous equations, a considerable amount of work went into the theory of determinants in the nineteenth century, and some elegant results that are very far from obvious were obtained. In the twentieth century interest moved from determinants to matrices partly due to swings of fashion in pure mathematics; matrices lend themselves much more easily to abstraction than determinants. More importantly from the point of view of applied mathematicians such as economists computers took over the nitty gritty job of calculating determinants and inverting matrices, so it became less important to know a lot about determinants, apart from the fact that they exist, and that tell you whether a matrix has an inverse. This chapter summarizes the facts about determinants that I think you should be aware of, but does not provide any proofs.

4.2 Notation and Definition

Only square matrices have determinants. The notation for the determinant of an $n \times n$ matrix

$$A = \begin{pmatrix} A_{11} & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} \end{pmatrix}$$

is

$$\det A = |A| = \begin{vmatrix} A_{11} & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} \end{vmatrix}.$$

The difference between notation for a determinant and notation for a matrix is that the determinant has lines $|.|$ where the matrix has brackets $(.)$.

If A is a 1×1 matrix (A_{11})

$$\det A = A_{11}.$$

If A is a 2×2 matrix

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

$$\det A = \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} = A_{11}A_{22} - A_{12}A_{21}.$$

You may find it helpful to remember this as the difference between the product of the diagonal terms and the product of the off diagonal terms.

The determinant of a 3×3 matrix A is

$$\det A = \begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{12} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{13} \begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix} \tag{4.1}$$

The determinant of an $n \times n$ matrix

$$A = \begin{pmatrix} A_{11} & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} \end{pmatrix}$$

is

$$\begin{aligned} \det A &= A_{11}M_{11} - A_{12}M_{12} + A_{13}M_{13} - ...... + (-1)^{1+c} M_{1c}.... + (-1)^{1+n} M \qquad (4.2) \\ &= \sum_{c=1}^{n} (-1)^{1+c} A_{1c}M_{1c} \end{aligned}$$

where M_{1c} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row 1 and column c from the matrix A.

$$M_{1c} = \begin{vmatrix} A_{21} & . & A_{2(c-1)} & A_{2(c+1)} & . & A_{2n} \\ A_{31} & . & A_{3(c-1)} & A_{3(c+1)} & . & A_{3n} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ A_{n1} & . & A_{n(c-1)} & A_{n(c+1)} & . & A_{nn} \end{vmatrix}$$

The $1c$ **cofactor** C_{rc} of the matrix A is

$$C_{1c} = (-1)^{1+c} M_{1c}.$$

Thus equation 4.2 can also be written as

$$\begin{aligned} \det A &= A_{11}C_{11} + A_{12}C_{22} + A_{13}C_{13} + ... + A_{1n-1}C_{1n-1} + A_{1n}C_{1n} \\ &= \sum_{c=1}^{n} A_{1c}C_{1c}. \end{aligned}$$

Note that only square matrices have determinants, so you should assume that all the matrices that you meet in this chapter are square.

4.3 Simultaneous Equations and Determinants

4.3.1 The 2×2 case

Determinants can appear mysterious; they turn out to have many mathematical properties that are far from obvious, but why should anyone have thought them interesting in the first place? The reason is that determinants emerge very naturally from the search for solutions to linear simultaneous equations.

Consider the equations

$$A_{11}x_1 + A_{12}x_2 = b_1 \qquad (4.3)$$

$$A_{21}x_1 + A_{22}x_2 = b_2. \qquad (4.4)$$

You will have learnt in high school mathematics that you can solve simultaneous equations either by substitution or by elimination of variables. The first step in substitution uses equation 4.4 to get x_2 as a function of x_1

$$x_2 = \frac{b_2 - A_{21}x_1}{A_{22}}. \qquad (4.5)$$

You then substitute this expression in equation 4.3 to get

$$A_{11}x_1 + A_{12}\left(\frac{b_2 - A_{21}x_1}{A_{22}}\right) = b_1$$

and rearrange to get

$$x_1 = \frac{b_1 - \left(\frac{A_{12}}{A_{22}}\right) b_2}{\left(A_{11} - \frac{A_{12}A_{21}}{A_{22}}\right)} = \frac{A_{22}b_1 - A_{12}b_2}{A_{11}A_{22} - A_{12}A_{21}}.$$

You can then substitute this expression for x_1 in equation 4.5 to get an expression

$$x_2 = \left(\frac{1}{A_{22}}\right)(b_2 - A_{21}x_1) = \frac{1}{A_{22}}\left(b_2 - A_{21}\left(\frac{A_{22}b_1 - A_{12}b_2}{A_{11}A_{22} - A_{12}A_{21}}\right)\right)$$

that after a bit of algebra reduces to

$$x_2 = \frac{b_2A_{11} - A_{21}b_1}{A_{11}A_{22} - A_{12}A_{21}}.$$

However the algebra is not much fun, and more seriously if either or both of A_{22} and $A_{11}A_{22} - A_{12}A_{21}$ are zero substitution involves the forbidden act of dividing by zero. If $A_{22} = 0$ equation 4.4 reduces to $A_{21}x_1 = b_2$ which provided $A_{21} \neq 0$ you can solve for $x_1 = b_2/A_{21}$. However if $A_{11}A_{22} - A_{12}A_{21} = 0$ trying to solve the equations by substitution runs into real difficulties. So if you are after a general solution it is better to work by elimination of variables, which avoids doing any division until the very last step.

The first step in solving the equations 4.3 and 4.4 by elimination of variables is to multiply equation 4.3 by A_{22} and equation 4.4 by A_{12} to get

$$\begin{aligned} A_{11}A_{22}x_1 + A_{12}A_{22}x_2 &= A_{22}b_1 \\ A_{12}A_{21}x_1 + A_{12}A_{22}x_2 &= A_{12}b_2. \end{aligned}$$

Subtracting the second of these equations from the first gives

$$(A_{11}A_{22} - A_{12}A_{21})\, x_1 = (A_{22}b_1 - A_{12}b_2) \tag{4.6}$$

Similarly multiplying equation 4.3 by A_{21} and equation 4.4 by A_{11} gives

$$\begin{aligned} A_{21}A_{11}x_1 + A_{12}A_{21}x_2 &= A_{21}b_1 \\ A_{21}A_{11}x_1 + A_{11}A_{22}x_2 &= A_{11}b_2. \end{aligned}$$

Subtracting the first of these equations from the second gives

$$(A_{11}A_{22} - A_{12}A_{21})\, x_2 = A_{11}b_2 - A_{21}b_1. \tag{4.7}$$

Writing equations 4.6 and 4.7 in determinant notation gives the result

$$(\det A)\, x_1 = \begin{vmatrix} b_1 & A_{12} \\ b_2 & A_{22} \end{vmatrix}. \tag{4.8}$$

$$(\det A)\, x_2 = \begin{vmatrix} A_{11} & b_1 \\ A_{21} & b_2 \end{vmatrix}. \tag{4.9}$$

This tells you two things. Firstly if $\det A = 0$ the only values of b_1 and b_2 for which there is a solution are those for which the right hand side of equations

4.8 and 4.9 are zero. Secondly if $\det A \neq 0$ you have a formula for the solution. Remember that

$$\det A = \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}$$

so the formula is

$$\begin{aligned} x_1 &= \frac{\begin{vmatrix} b_1 & A_{12} \\ b_2 & A_{22} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ x_2 &= \frac{\begin{vmatrix} A_{11} & b_1 \\ A_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}}. \end{aligned}$$

The first column of the matrix inverse is the solutions to the equations

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 &= b_1 \\ A_{21}x_1 + A_{22}x_2 &= b_2. \end{aligned}$$

with $b_1 = 1$ and $b_2 = 0$. The second column of the inverse is the solution to the equations with $b_1 = 0$, and $b_2 = 1$. This gives the components of A^{-1} as

$$\begin{aligned} A^{-1}_{11} &= \frac{\begin{vmatrix} 1 & A_{12} \\ 0 & A_{22} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{A_{22}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ A^{-1}_{21} &= \frac{\begin{vmatrix} A_{11} & 1 \\ A_{21} & 0 \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{-A_{21}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ A^{-1}_{12} &= \frac{\begin{vmatrix} 0 & A_{12} \\ 1 & A_{22} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{-A_{12}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ A^{-1}_{22} &= \frac{\begin{vmatrix} A_{11} & 0 \\ A_{21} & 1 \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{A_{11}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}}. \end{aligned}$$

I have to admit that I find these formulae deeply unmemorable, my memory works better with words:

- The determinant of a 2×2 matrix is the product of its diagonal terms minus the product of the off diagonal terms.
- The inverse of a 2×2 matrix is found by swapping the diagonal terms, changing the sign of the off diagonal terms and the dividing by the determinant.

4.4 Determinants, Inverses and Cramer's Rule

These results generalize from the 2×2 to the $n \times n$ case. The most important result on determinant is:

Theorem 20 (Determinants and Matrix Inverses) *The $n \times n$ matrix has an inverse if and only if* $\det A \neq 0$. *If* $\det A \neq 0$ *the the inverse of A is*

$$A^{-1} = \left(\frac{1}{\det A} \right) \operatorname{adj} A$$

where $\operatorname{adj} A$ *is the* ***adjoint*** *of A, that is the matrix whose ij entry is $(-1)^{i+j} M_{ji}$ where M_{ji} is the ji minor, that is the determinant of the matrix formed by deleting row j and column i from the matrix A.*

Note in the last part of this proposition that the ij component of $\operatorname{adj} A$, that is the entry in **row i** and **column j** of $\operatorname{adj} A$, is the determinant of the matrix formed by deleting **row j** and **column i** from A. This is not a typo.

This result makes it possible to solve the equation $Ax = b$ when $\det A \neq 0$ by using the formula in Theorem 20 to find A^{-1} and thus find the solution $x = A^{-1}b$. There is an alternative approach, **Cramer's Rule**, that makes direct use of determinants:

Theorem 21 (Cramer's Rule) *If $\det A \neq 0$ the equation $Ax = b$ has a unique solution for every b. The solution is*

$$x_i = \frac{\det B_i}{\det A} \qquad i = 1, 2....n$$

where B_i is the $n \times n$ matrix obtained by replacing the i^{th} column of A by the vector b.

Cramer's rule is used in economics, however if n is large calculating the determinants is burdensome, and if n is small it is often easier to solve the equations directly by substitution or elimination of variables, or equivalently using row echelon matrices.

4.5 Calculating Determinants: Some Special Cases

4.5.1 The 3×3 Case

It is possible in principle to calculate the determinant of any matrix working with the definition. This is easy for a 2×2 matrix, and not bad for a 3×3 matrix where the definition implies that

$$\begin{aligned}
&\begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} \\
= \; & A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{12} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{13} \begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix} \\
= \; & A_{11}A_{22}A_{33} + A_{12}A_{23}A_{31} + A_{13}A_{21}A_{32} \\
& -A_{11}A_{23}A_{32} - A_{12}A_{21}A_{33} - A_{13}A_{22}A_{31}.
\end{aligned}$$

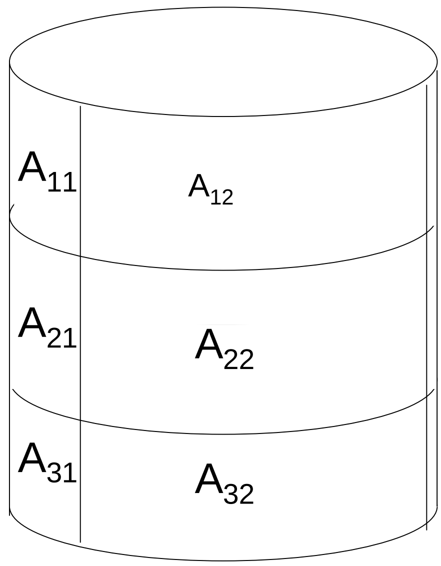

Figure 4.1:

I think of this formula by visualizing the matrix pasted on a cylinder as in Figure 4.1. The terms with a plus sign $+$ come from taking an element on the top row of the cylinder and then finding two more elements by moving one place to the right and one place down twice. The terms with a negative sign $-$ come from taking an element on the top row and then finding two more elements by moving one place to the left and one place down twice. Unfortunately this visualization trick only works for 2×2 and 3×3 matrices.

4.5.2 Diagonal Matrices

Definition 22 *The $n \times n$ matrix A is **diagonal** if all the off diagonal terms are zero, that is $A_{ij} = 0$ when $i \neq j$.*

For example the matrix

$$\begin{pmatrix} A_{11} & 0 & 0 \\ 0 & A_{22} & 0 \\ 0 & 0 & A_{33} \end{pmatrix}$$

is diagonal. There is an easy result on the formula for the determinant of a diagonal matrix.

Theorem 23 (Determinants of Diagonal Matrices) *The determinant of a diagonal matrix is the product of its diagonal terms.*

To see why this is so let

$$A = \begin{pmatrix} A_{11} & 0 & . & . & \\ 0 & A_{22} & 0 & . & \\ . & 0 & A_{33} & 0 & \\ . & . & .0 & . & 0 \\ . & . & . & 0 & A_{nn} \end{pmatrix}.$$

Then

$$\det A = A_{11} \begin{vmatrix} A_{22} & 0 & . & \\ 0 & A_{33} & 0 & \\ . & .0 & . & 0 \\ . & . & 0 & A_{nn} \end{vmatrix} = A_{11}A_{22} \begin{vmatrix} A_{33} & . & . \\ . & . & . \\ . & . & A_{nn} \end{vmatrix} = A_{11}A_{22}.....A_{nn}.$$

4.5.3 The Identity Matrix

The identity matrix I is a diagonal matrix for which all the diagonal terms are 1. The result on diagonal matrices implies at once:

Proposition 24 *The determinant of the identity matrix is* 1.

4.6 The General Case

4.6.1 Use a Computer

Calculating the determinant of a general $n \times n$ matrix with $n > 3$ by hand is usually a deeply tedious piece of algebra or arithmetic with a huge risk of making mistakes. This is a job for a computer, as is finding the inverse of a matrix. Mine has just told me in seconds that the matrix

$$\begin{pmatrix} -5 & 3 & -2 & 4 \\ -2 & 1 & -1 & 2 \\ 4 & -2 & 4 & 6 \\ -7 & 4 & -3 & 11 \end{pmatrix}$$

has determinant 10 and inverse

$$\begin{pmatrix} -0.4 & -5.4 & -0.5 & 1.4 \\ 0.6 & -7.4 & -0.5 & 1.4 \\ 1.0 & 2.0 & 0.5 & -1.0 \\ -0.2 & -0.2 & 0.0 & 0.2 \end{pmatrix}.$$

4.7 Expanding Along Rows and Columns: optional

Look quickly at this material

4.7.1 The Expansion Result

If you do have to calculate a determinant by hand there are some results that may make the calculation easier which derive from the result on expanding along rows and columns which I am about to state.

Equation 4.2 gives the general formula for the determinant of an $n \times n$ matrix

$$\begin{aligned} \det A &= A_{11}M_{11} - A_{12}M_{12} + A_{13}M_{13} - \ldots\ldots \\ &\quad + (-1)^{1+c} A_{1c}M_{1c} \ldots. + (-1)^{1+n} A_{1n}M \\ &= \sum_{c=1}^{n} (-1)^{1+c} A_{1c}M_{1c} \end{aligned}$$

where M_{1c} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row 1 and column c from the matrix A. M_{rc} is called the $1c$ **minor** of the matrix A. I think of this as expanding along row 1, and remember the formula as the sum over $c = 1, 2 \ldots n$ of terms of the form

element c of row 1
$\times (-1)^{1+c}$
$\times$determinant of the matrix formed from A by deleting row 1 and column c.

It is in fact possible to expand along any row or column. Stating this result uses the general definition of a minor.

Definition 25 *The rc minor M_{rc} of the matrix A is the determinant of the $(n-1)\times(n-1)$ matrix obtained by deleting row r and column c from the matrix A.*

The formal result on expanding along a row or column is:

Theorem 26 (Expansion of a Determinant Along a Row or Column) *If A is an $n \times n$ matrix then for any row r*

$$\det A = \sum_{c=1}^{n} (-1)^{r+c} A_{rc} M_{rc} \tag{4.10}$$

and for any column c

$$\det A = \sum_{r=1}^{n} (-1)^{r+c} A_{rc} M_{rc}. \tag{4.11}$$

It is straightforward to verify the result for a 2×2 matrix where $M_{11} = A_{22}$, $M_{12} = A_{21}$, $M_{21} = A_{12}$ and $M_{22} = A_{11}$.

I think of equation 4.10 as expanding along row r, remembering that $\det A$ is the sum over $c = 1, 2...n$ of terms

element c of row r
$\times (-1)^{r+c}$
$\times$determinant of the matrix formed from A by deleting row r and column c.

I think of equation 4.11 as expanding along column c, remembering that $\det A$ is the sum over $r = 1, 2...n$ of terms

element r of column c
$\times (-1)^{r+c}$
$\times$determinant of the matrix formed from A by deleting row r and column c.

An example using this result is:

Example 27 *Expanding along column 2 of the matrix*

$$\begin{pmatrix} A_{11} & b_1 & A_{13} \\ A_{21} & b_2 & A_{23} \\ A_{31} & b_3 & A_{33} \end{pmatrix}$$

gives

$$\begin{vmatrix} A_{11} & b_1 & A_{13} \\ A_{21} & b_2 & A_{23} \\ A_{31} & b_3 & A_{33} \end{vmatrix} = -b_1 \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + b_2 \begin{vmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{vmatrix} - b_3 \begin{vmatrix} A_{11} & A_{13} \\ A_{21} & A_{23} \end{vmatrix}.$$

Proposition 26 is particularly useful if you are working with a matrix that has several zeros in the same row or column. For example, expanding along the bottom row

$$\begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ 0 & A_{32} & 0 \end{vmatrix} = -A_{32} \begin{vmatrix} A_{11} & A_{13} \\ A_{21} & A_{23} \end{vmatrix} = -A_{32} \left(A_{11}A_{23} - A_{13}A_{21} \right).$$

4.7.2 Consequences of the Expansion Result

The Determinant of a Matrix with a Row or Column of Zeros is Zero

This is a list of results that are sometimes useful and follow easily from the expansion result Theorem 26. The first set of results can all be proved very simply by expanding along the relevant row or column. The first of these is particularly worth remembering.

Proposition 28 • *If all the elements of a row of a square matrix* A *are* 0 *then* $\det A = 0$.

- *If all the elements of a column of a square matrix* A *are* 0 *then* $\det A = 0$.

Determinants of Triangular Matrices are the Product of the Diagonal Terms

Two more results that follow quite straight forwardly from the expansion result; these involve upper and lower triangular matrices.

Definition 29 *The* $n \times n$ *matrix* A *is* ***upper triangular*** *if all the terms below the diagonal are zero, that is* $A_{ij} = 0$ *when* $i > j$.

For example the matrix

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ 0 & A_{22} & A_{23} \\ 0 & 0 & A_{33} \end{pmatrix}$$

is upper triangular.

Proposition 30 *The determinant of an upper triangular matrix is the product of its diagonal terms.*

The result follow quite easily by expanding along column 1 the determinant of an $n \times n$ upper triangular matrix

$$\begin{aligned} \begin{vmatrix} A_{11} & A_{12} & A_{13} & . & . & A_{1n} \\ 0 & A_{22} & A_{23} & . & . & A_{2n} \\ 0 & 0 & A_{33} & . & . & . \\ 0 & 0 & 0 & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & 0 & 0. & 0 & A_{nn} \end{vmatrix} &= A_{11} \begin{vmatrix} A_{22} & A_{22} & . & . & A_{2n} \\ 0 & A_{33} & . & . & A_{3n}. \\ 0 & 0 & . & . & . \\ . & . & . & . & . \\ 0 & 0 & 0. & 0 & A_{nn} \end{vmatrix} \\ &= A_{11}A_{22} \begin{vmatrix} A_{33} & . & . & A_{3n}. \\ 0 & . & . & . \\ . & . & . & . \\ 0 & 0. & 0 & A_{nn} \end{vmatrix} \\ &= A_{11}A_{22}A_{33}......A_{nn}. \end{aligned}$$

Definition 31 *The* $n \times n$ *matrix* A *is* ***lower triangular*** *if all the terms above the diagonal are zero, that is* $A_{ij} = 0$ *when* $i < j$.

For example the matrix

$$\begin{pmatrix} A_{11} & 0 & 0 \\ A_{21} & A_{22} & 0 \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$$

is lower triangular.

Proposition 32 *The determinant of a lower triangular matrix is the product of its diagonal terms.*

The proof is very similar to the corresponding proposition for an upper triangular matrix, except the determinant is expanded along the first row.

4.8 Determinants of Related Matrices

The most important result here is

Theorem 33 *Determinants and Products. The determinant of the product of two $n \times n$ matrices A and B is the product of the determinants, that is*

$$|AB| = |A| \, |B| \, .$$

As $\det I = 1$ this implies that if A has an inverse A^{-1} then $(\det A) \left(\det A^{-1}\right) = \det \left(AA^{-1}\right) = \det I = 1$. If $\det A = 0$ this implies the impossible $0 = 1$, and thus proves:

Theorem 34 *If A has an inverse then $\det A \neq 0$ and*

$$\det \left(A^{-1}\right) = \frac{1}{\det A}.$$

There is also a useful result on the transpose A' of the matrix A. (The transpose A' of A is the matrix in whose row vectors are the column vectors of A, so $A'_{ij} = A_{ji}$). The result is

Proposition 35 *The determinant of a matrix A and the determinant of its transpose A' are equal.*

$$\det A = \det \left(A'\right).$$